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ADDED SUGAR THE FACT-CHECK

WHAT IT IS, HOW MUCH WE EAT, AND WHAT THE SCIENCE REALLY SAYS

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Before You Read

Educational information only. Not medical advice. This paper explains what public health agencies and published research say about added sugar. It does not diagnose, treat, or promise any result. Talk with your doctor or a registered dietitian before you make big changes to how you eat, **especially if you have diabetes or take medication that lowers blood sugar.** Your care team can help you make changes safely.

How to use this paper:

- **Short on time?** Jump to the **Cheat Sheet** on the last pages before the references.
- **Want the truth about a claim you heard?** Go to **Myth vs. Fact**. Every claim gets a verdict.
- **Want to check our work?** Every number has a small bracket like this: [1]. Match it to the numbered **References** at the end. Every source is public and checkable.

Our verdict scale:

Verdict	What it means
True	The evidence clearly supports the claim.
Mostly true	Right in the main, but missing an important detail.
Misleading	Has a grain of truth, but leads people to the wrong conclusion.
It's complicated	The honest answer depends on the details, or the science is still unsettled.
False	The evidence does not support the claim.

How strong is the evidence? We tell you that too. Here is the short version:

- **Randomized trials** split people into groups by chance and change one thing. They are the best way to show cause and effect. But they are often short and small.
- **Observational studies** follow large groups of people over years and look for patterns. They can show that two things are **linked**. They cannot prove one **causes** the other on their own.
- **Animal studies** help explain how something might work. They do not prove what happens in people.

When we say sugar is "**linked to**" a health problem, that is the careful, honest way to describe observational research. We will tell you when the evidence is stronger than that.

The 60-Second Version

- **Added sugar** is sugar put into food or drinks during processing or prep. It is not the sugar that comes packed inside whole fruit or plain milk [1].
- **The average American adult eats about 17 teaspoons of added sugar a day.** So do kids and teens ages 2 to 19 [5].
- **Sugary drinks are the #1 source.** They make up about **24%** of added sugar in the U.S. diet [4].
- **One 20-ounce Coca-Cola has 65 grams of added sugar** [11]. That is **about 16 teaspoons**, and **130%** of the Daily Value on the label.
- **The newest U.S. Dietary Guidelines (2025–2030, released January 7, 2026)** say no amount of added sugar is recommended as part of a healthy diet. As a practical limit, they say **one meal should have no more than 10 grams of added sugar** [3].
- **Kids:** the 2025–2030 Guidelines say to avoid added sugars from birth through age 4, and that no amount is recommended for ages 5 to 10 [3].
- Eating a lot of added sugar, especially from drinks, is **linked to** weight gain, cavities, type 2 diabetes, heart disease, and fatty liver [4][10][21][34][38].
- **Whole fruit is not the problem.** Juice and sugary drinks are a different story [22][23].
- **The fix is simple, not easy:** read the label, drink water, and cut the sweet drinks first.

Part 1: What Counts as "Added Sugar"?

Two kinds of sugar

Sugar shows up in food in two ways.

1. Naturally occurring sugar. This sugar is built into the food. Think of the sugar in an apple, a banana, a carrot, or a glass of plain milk. It comes packed with water, fiber, protein, vitamins, or minerals. The FDA does **not** count this as added sugar [1].

2. Added sugar. This is sugar that someone put in. The FDA's definition includes [1]:

- Sugars added during processing (like sucrose or dextrose)
- Foods sold as sweeteners (like a bag of table sugar)
- Sugars from syrups and honey
- Sugars from concentrated fruit or vegetable juices

Simple test: If a person or a factory put the sweet stuff in, it is added sugar. If it grew that way, it is not.

Your body uses both the same way, so why the difference?

Fair question. Once digested, your body handles the sugar molecules the same way [12]. The difference is **the package**.

The sugar in an apple comes with fiber and takes time to chew and digest. The American Heart Association points out that your body spends more time digesting an apple than a soda. The sugar in soda "hits your system all at once" [7].

Harvard Health notes that for most people, natural sugar in foods like fruit is not linked to negative health effects. Our bodies do not need, or benefit from, added sugar [12].

The "Added Sugars" line on the Nutrition Facts label

The FDA updated the Nutrition Facts label. Big food companies (those with **\$10 million or more** in yearly food sales) had to switch by **January 1, 2020**. Smaller companies had until **January 1, 2021** [2].

The biggest change for sugar lovers: a new line that says **"Includes Xg Added Sugars."** It sits right under "Total Sugars."

Here is how to read it, using the FDA's own yogurt example [1]:

Label line	What it says	What it means
Total Sugars	15 g	All sugar in one serving, natural + added
Includes Added Sugars	7 g	7 g of the 15 g was put in
% Daily Value (Added Sugars)	14%	7 g is 14% of the 50 g daily limit
Natural sugar (do the math)	8 g	$15 - 7 = 8$ g came from the milk itself

The Daily Value for added sugars is 50 grams per day, based on a 2,000-calorie diet [1]. That number came from the 2020–2025 Dietary Guidelines' limit of less than 10% of calories from added sugar. 10% of 2,000 calories is 200 calories, or 50 grams [1].

The FDA's rule of thumb: 5% DV or less of added sugars is **LOW**. **20% DV or more** is **HIGH** [1].

One quirk: Packages of honey, maple syrup, or plain table sugar are labeled a little differently. They show the %DV for added sugars, and the gram amount may appear in a footnote. The FDA does this so it does not look like sugar was "added" to sugar [1]. These are still added sugars when you put them in your food.

Another note: "Total Sugars" has **no** Daily Value. Experts have not set a daily target for total sugar [1]. The focus is on the added kind.

Sugar by the teaspoon: the quick math

Most of us cannot picture "65 grams." We can picture teaspoons.

1 teaspoon of sugar $\approx$ 4 grams. The FDA uses "two teaspoons (approximately 8 grams) of sugar" in its own comparisons [29]. The AHA's limits use the same math: 25 g is about 6 teaspoons [7].

To convert: grams on the label $\div$ 4 = teaspoons.

Calories: each gram of sugar has about 4 calories [7]. 25 g of sugar = about 100 calories.

Part 2: Sugar Has a Lot of Aliases

Food companies do not have to call it "sugar." The ingredient list may use dozens of names. The newest Dietary Guidelines give a simple trick [3]:

Look for ingredients that include the word "sugar" or "syrup," or that end in "-ose."

Here are common names, pulled from the 2025–2030 Dietary Guidelines [3], the CDC [6], and the American Heart Association [7][8]:

Group	Names you might see
"Sugar" in the name	Sugar, cane sugar, beet sugar, brown sugar, raw sugar, turbinado sugar, coconut sugar, sugar cane
Syrups	High fructose corn syrup (HFCS), corn syrup, rice syrup, malt syrup, maple syrup, agave syrup
Ends in "-ose"	Sucrose, glucose, dextrose (a form of glucose), fructose, maltose, lactose*
"Natural-sounding" sweeteners	Honey, molasses, agave nectar, fruit nectars
Juice tricks	Fruit juice concentrate
Other	Corn sweetener

Lactose is the natural sugar in milk. In plain milk it is **not added sugar. When lactose is added as a separate ingredient, it is [3].*

Why so many names matters

Ingredients are listed from most to least by weight. If a product uses three or four different sweeteners, each one may sit lower on the list. That can hide how much sugar is really there.

Do not play detective with the ingredient list alone. Since 2020, the **Added Sugars** line does the math for you [1][2]. Use the ingredient list to see *what kind* of sweetener. Use the Added Sugars line to see *how much*.

What about "no-calorie" sweeteners?

Aspartame, sucralose, saccharin, acesulfame K, and stevia are **not** sugars. They are **non-nutritive sweeteners** [3][29]. They are not counted on the Added Sugars line. We cover them in Myth #11.

The 2025–2030 Dietary Guidelines list xylitol with these sweeteners [3]. Technically, xylitol is a sugar alcohol. Sugar alcohols have a few calories [29].

Part 3: How Much Sugar Do Americans Really Eat?

The big picture

Here is what the latest federal data show.

Who	Average added sugar per day	Source
Adults 20+	17 teaspoons (about 68 g)	CDC, NHANES 2017–2018 [5]
Men 20+	19 teaspoons	CDC [5]
Women 20+	15 teaspoons	CDC [5]
Kids and teens 2–19	17 teaspoons	CDC, NHANES 2017–2018 [5]
Boys 2–19	18 teaspoons	CDC [5]
Girls 2–19	15 teaspoons	CDC [5]
Everyone 1 and older	About 270 calories a day , more than 13% of all calories	2020–2025 Dietary Guidelines [4]

Grams are our conversion at about 4 g per teaspoon.

Put another way: the **average** adult eats almost **three times** the AHA's limit for women (6 teaspoons) and about **twice** the limit for men (9 teaspoons) [5][7].

Not just a few people. CDC research found about **30% of Americans age 2 and older** get **more than 15% of their daily calories** from added sugar [6].

Kids and teens: the numbers climb with age

The CDC breaks down kids' intake by age and background [5]:

Age group	Average added sugar per day (range across groups)
Ages 2–5	7 to 13 teaspoons
Ages 6–11	12 to 19 teaspoons
Ages 12–19	14 to 20 teaspoons

The 2020–2025 Dietary Guidelines also show how many kids go over the old limit of 10% of calories [4]:

Age group	Boys over the limit	Girls over the limit
2–4	61%	57%
5–8	80%	77%
9–13	79%	78%
14–18	72%	76%

A few more facts that stopped us cold:

- **Toddlers** (ages 1 to 2) take in an average of **more than 100 calories** from added sugars a day. Sugary drinks, mostly fruit drinks, supply more than **25%** of it [4].
- Added sugar **peaks at about 15% of calories in the teen years** [4].
- For teens, sugary drinks supply about **32%** of added sugar. Sweetened coffee and tea add another **7%** [4].
- From 2021 to 2023, **55.9% of children ages 1 to 5** had at least one sugary drink in the past week [5].

Adults

Under the 2020–2025 Guidelines, **about 60% of men and 65% of women** went over the added sugar limit. That averaged about **330 calories a day for men** and about **250 for women** [4].

A note on "latest" data. The CDC's most recent published added sugar intake numbers come from the 2017–2018 national survey [5]. A newer survey (August 2021–August 2023) has been reported for **ultra-processed foods**. It found that Americans age 1 and older got **55%** of their calories from ultra-processed foods (youth **61.9%**, adults **53.0%**) [39]. Many of those foods are major sources of added sugar.

Where does it all come from?

Most added sugar does not come from the sugar bowl. It comes from drinks and packaged foods. Here is the breakdown for Americans age 1 and older [4]:

Source	Share of all added sugar
Sugar-sweetened beverages (soda, fruit drinks, sports & energy drinks)	24%
Desserts & sweet snacks (cookies, cakes, ice cream, pastries)	19%
Coffee & tea (with sugar added)	11%
Candy & sugars	9%
Breakfast cereals & bars	7%
Sandwiches	7%
Higher-fat milk & yogurt (sweetened types)	4%
Everything else	19%

Data: What We Eat in America, NHANES 2013–2016. Soft drinks alone = 16% [4].

The top four (drinks, desserts, sweetened coffee and tea, and candy) make up more than half of all added sugar, and give "very little" toward food group needs [4].

What's in a glass? Common drinks, 12 ounces each

The CDC lists the sugar in common 12-ounce drinks, based on USDA data [6]:

Drink (12 oz)	Teaspoons of sugar	Calories
Plain water	0	0
Unsweetened tea	0	0
Lemonade (powder, mixed with water)	3	55
Sports drink	5	97
Brewed sweet tea	7	115
Energy drink	9	162
Regular soda	10	155
Fruit juice drink	10	186
Regular orange soda	13	195

The 20-ounce bottle check. Coca-Cola's own nutrition facts: a **12 oz can has 39 g** of sugar (about **10 teaspoons**). A **20 oz bottle has 65 g** of sugar and 240 calories (about **16 teaspoons**) [11]. The label lists that 65 g as **added sugars, 130% of the Daily Value** [11]. That one bottle is more than **2½ times** the AHA's daily limit for women and kids [7][9].

Part 4: How Much Is Too Much? The Official Limits

Different expert groups set different limits. They all point the same way: **less**.

The newest U.S. guidance: Dietary Guidelines for Americans, 2025–2030

Released January 7, 2026 by the U.S. Department of Agriculture (USDA) and the Department of Health and Human Services (HHS) [3]. This is now the current version. Here is what it says about sugar, in its own terms [3]:

- "No amount of added sugars or non-nutritive sweeteners is recommended or considered part of a healthy or nutritious diet."
- "One meal should contain no more than **10 grams** of added sugars." (That's about **2½ teaspoons**.)
- Avoid sugar-sweetened beverages, such as sodas, fruit drinks, and energy drinks.
- For snacks, follow FDA "Healthy" claim limits. Example: grain snacks like crackers should not go over **5 g** of added sugar per ¾-ounce whole-grain serving. Dairy snacks like yogurt should not go over **2.5 g** per ⅔-cup serving.
- **Infancy and early childhood (birth–4 years):** avoid added sugars.
- **Middle childhood (5–10 years):** "No amount of added sugars is recommended."
- **Teens (11–18):** significantly limit sugary drinks and energy drinks.
- 100% fruit or vegetable juice should be "consumed in limited portions or diluted with water."

What changed? The 2020–2025 version capped added sugar at **less than 10% of calories** for ages 2 and up, and said to avoid it **under age 2** [4]. Harvard's Nutrition Source calls the new 10-gram-per-meal limit a reduction from that old 10% cap. It also notes the new guidelines move the "avoid added sugars" age for kids from 2 up to 10 [40]. Harvard also points out that most people do not track sugar meal by meal, so putting it into practice may be tricky [40].

Heads up on the label: The FDA's **50 g Daily Value** is still based on the older 10% rule [1]. So a food can show a "low" %DV and still go over the new **10 g per meal** benchmark if you eat a few servings. Use both numbers.

The American Heart Association (AHA)

The AHA recommends keeping added sugars **below 6% of calories per day** [7]:

- **Most men:** no more than **9 teaspoons (36 grams, 150 calories)** a day [7].
- **Most women:** no more than **6 teaspoons (25 grams, 100 calories)** a day [7].

- **Kids and teens ages 2–18: 25 grams (about 6 teaspoons) or less** a day [9].
- **Kids under 2:** avoid added sugars [9].
- **Kids and teens:** no more than **one 8-ounce sugary drink per week** [9].

The World Health Organization (WHO)

The WHO's 2015 guideline is about "**free sugars**." That is added sugar **plus** sugar in honey, syrups, and fruit juice [10].

- **Strong recommendation:** keep free sugars **below 10%** of daily calories, for adults and children [10].
- **Conditional recommendation:** go further, to **below 5%**, or **roughly 25 grams (6 teaspoons)** a day, for extra benefit [10].

The WHO is honest about its evidence. The 10% advice rests on moderate-quality evidence. The 5% advice rests on very low-quality evidence, mostly about tooth decay [10].

Side-by-side table (for a 2,000-calorie day)

Sources: [1][3][4][5][7][9][10]. Teaspoons converted at about 4 g per teaspoon; the AHA and WHO 25 g figures are their own.

Guidance	Limit	Grams	Teaspoons
DGA 2025–2030 (current)	No amount recommended; ≤10 g per meal	≤10 g per meal	≈2½ per meal
DGA 2025–2030, kids birth–10	Avoid / none recommended	0	0
DGA 2020–2025 (previous)	<10% of calories (age 2+)	<50 g	<12½
FDA label Daily Value	50 g	50 g	≈12½
AHA, most women	≤6% of calories	≤25 g	≤6
AHA, most men	≤6% of calories	≤36 g	≤9
AHA, kids 2–18	≤25 g; ≤1 sugary drink (8 oz) per week	≤25 g	≤6
WHO (free sugars)	<10% of calories	<50 g	<12½
WHO, extra benefit	<5% of calories	≈25 g	≈6
Average U.S. adult actually eats	—	≈68 g	17

Part 5: Myth vs. Fact

Twelve things people say about sugar. Here is what the evidence actually shows.

Myth #1: "Sugar is as addictive as cocaine."

VERDICT: MISLEADING

Where it comes from: A 2007 study in rats. When rats could choose between sweetened water and an IV dose of cocaine, **94%** picked the sweet water (sweetened with saccharin). They showed the same preference with real sugar [16].

What the evidence says: That is a striking rat study. But rats are not people. A 2016 review looked at the human and animal research on "sugar addiction." The authors found **little evidence to support sugar addiction in humans**. In animals, binge-like behavior showed up mostly when sugar was given **on and off** (intermittent access), not all the time. They argued against treating sugar addiction as settled science [15].

The grain of truth: Sweet foods are very rewarding. Many people find them hard to stop eating. That feeling is real. It just is not the same thing as a drug addiction.

Evidence strength: Animal studies are interesting. Human evidence is weak.

Bottom line: Sugar is easy to overdo. Calling it "cocaine" overstates the science.

Myth #2: "Sugar makes kids hyper."

VERDICT: FALSE (for most kids)

What the evidence says: A 1995 analysis in *JAMA* pooled 23 studies where neither the kids, parents, nor researchers knew who got sugar and who got a sugar-free placebo. The result: **sugar did not affect the behavior or thinking of children**. The authors said a small effect, or an effect in some kids, can't be fully ruled out [17].

So why do so many parents see it? A clever 1994 study gave **every** boy a sugar-free drink. Then researchers told half the moms their sons had eaten a lot of sugar. Those moms **rated their sons as more hyperactive**. They also hovered and criticized more [18].

The grain of truth: Sugar often shows up at birthday parties, holidays, and late nights. The excitement is real. The sugar likely isn't the cause.

Evidence strength: Strong. These were blinded trials.

Bottom line: Sugar is not a behavior drug. But kids still don't need it: the 2025–2030 Guidelines recommend none for kids 10 and under [3].

Myth #3: "Sugar directly causes diabetes."

VERDICT: IT'S COMPLICATED

Type 1 diabetes: This is an autoimmune condition. Eating sugar does not cause it.

Type 2 diabetes: No single food "causes" it. The American Diabetes Association lists risk factors like weight, physical activity, family history, ethnicity, and age [19]. But the ADA also says research shows **sugary drinks are linked to type 2 diabetes**. It recommends avoiding them and switching to water [19].

What the research shows:

- A 2010 review of 11 studies (310,819 people) found those drinking the most sugary drinks (most often 1–2 a day) had a **26% higher risk** of type 2 diabetes than those drinking the least [20].
- A 2015 *BMJ* review found each daily serving of a sugary drink was linked to an **18% higher risk** of type 2 diabetes. After accounting for body weight, it was still **13% higher** [21]. That suggests weight is a big part of the story, but maybe not all of it.

Evidence strength: Strong and consistent, but observational. These studies show a link, not proof of direct cause.

Bottom line: Sugar isn't a switch that flips on diabetes. But a steady habit of sugary drinks is **linked to higher type 2 risk**, partly through weight gain.

Myth #4: "Honey, agave, and coconut sugar are 'healthy' sugars."

VERDICT: FALSE

What the evidence says: Your body handles these sugars the same way as other added sugars [12]. The FDA counts sugar from honey and syrups as **added sugar** [1]. The AHA puts it bluntly: "**Added sugar is added sugar no matter what it is called**" [7].

- **Honey** has about a 1-to-1 mix of fructose and glucose, like table sugar [13]. It has small amounts of other compounds, but Harvard Health says those "have little benefit when it comes to metabolic health" [12].
- **Agave** is marketed as natural and gentle. Some types are **up to 90% fructose** [12]. That does not make it healthier.

- **Coconut sugar** is still sugar. The AHA lists it as an added sugar to watch for [7].

Evidence strength: Strong. This is basic food chemistry plus federal labeling rules.

Bottom line: "Natural" is a marketing word. Count these just like white sugar.

Myth #5: "Fruit is just as bad as candy."

VERDICT: FALSE

What the evidence says: Whole fruit is a different food from candy. The sugar comes with fiber, water, and nutrients. The FDA does not count it as added sugar [1]. Every major guideline tells people to **eat** fruit. The 2025–2030 Guidelines set a goal of **2 servings of fruit a day** on a 2,000-calorie pattern and say to eat it "in [its] original form" [3].

A huge Harvard study followed more than 187,000 nurses and health professionals for years. **Whole fruit** was linked to a slightly **lower** risk of type 2 diabetes (blueberries, grapes, and apples stood out). **Fruit juice** was linked to a **higher** risk [22].

The grain of truth: Fruit **juice**, dried fruit with added sugar, and fruit "snacks" are not the same as whole fruit. Read the label on canned fruit in syrup.

Evidence strength: Good. Large, long-term observational studies agree with expert guidance.

Bottom line: Eat the apple. Skip the apple-flavored candy.

Myth #6: "High fructose corn syrup is way worse than table sugar."

VERDICT: MISLEADING

What the evidence says: HFCS and table sugar are close cousins.

- **Table sugar (sucrose)** is one glucose joined to one fructose: a **50/50** split [13].
- The most common forms of **HFCS** are **42% or 55% fructose**, with the rest glucose and water. HFCS-55 is used mostly in soft drinks [13].

The FDA says it is not aware of any evidence of a safety difference between HFCS and similar amounts of table sugar, honey, or other traditional sweeteners [13]. The AHA's limits cover all added sugars "without singling out a type such as high fructose corn syrup" [8].

The grain of truth: HFCS shows up in a lot of sodas and cheap processed foods. People who eat a lot of it are often eating a lot of added sugar overall. The problem is the **amount**, not the brand of sugar.

Evidence strength: Good, based on the chemistry and on FDA and AHA positions.

Bottom line: Swapping HFCS soda for "made with real cane sugar" soda doesn't fix much. Cutting the soda does.

Myth #7: "Fat-free foods are the healthier choice."

VERDICT: MISLEADING

What the evidence says: When companies take fat out, they often add sugar back in to keep the taste. A 2016 study compared regular, low-fat, and fat-free versions of the same foods using USDA data. **Low-fat and fat-free versions had more sugar** than the regular versions [24].

The grain of truth: Some lower-fat foods can fit a healthy pattern. Plain fat-free milk does not have added sugar. But "fat-free" on the front of a box says nothing about sugar. The 2025–2030 Guidelines now encourage **full-fat dairy with no added sugars** [3].

Evidence strength: Moderate. The 2016 study compared nutrient data, not health outcomes.

Bottom line: Ignore the front of the box. Flip it over and read **Added Sugars**.

Myth #8: "Brown sugar is healthier than white sugar."

VERDICT: FALSE

What the evidence says: Most brown sugar is white sugar with some molasses left in or added back. Brown sugar is usually at least **85% sucrose**. White sugar is about **99.7%** [14]. Both have about the same calories per teaspoon [14].

Yes, brown sugar has tiny amounts of minerals like calcium and iron. But you would need about **100 grams of brown sugar (20 teaspoons, about 300 calories)** to get just **6% of your daily calcium** or **4% of your daily iron** [14]. Turbinado and "raw" sugars are in the same boat [14].

Evidence strength: Strong. This is basic nutrient data.

Bottom line: Brown sugar tastes different. It is not better for you.

Myth #9: "A calorie is a calorie, so sugar doesn't matter."

VERDICT: MISLEADING

The grain of truth: Calories count. A 2013 *BMJ* review of trials found that when adults ate **less** sugar, they lost a little weight. When they ate **more**, they gained. But when sugar was swapped for **the same calories** of other carbs, weight did not change [30]. That tells us sugar's effect on weight works mostly through **extra calories**.

Why "sugar doesn't matter" is still wrong:

- **Liquid sugar sneaks past your fullness signals.** In a small study, people given 450 calories a day of soda did not eat less food to make up for it, and they gained weight. When given the same calories as jelly beans, they ate less at other times [25].
- **How food is made affects how much we eat.** In an NIH trial, people on an ultra-processed diet ate about **500 more calories a day** than on an unprocessed diet, even though the meals were matched for sugar, fat, fiber, and more [41].
- **Sugar crowds out nutrients.** The 2020–2025 Guidelines warn that too much added sugar makes it hard to get the nutrients you need without going over your calorie needs [1][4].
- **Teeth react to the sugar itself.** More free sugar is linked to more cavities [10][33].

Evidence strength: Good. Includes randomized trials.

Bottom line: Calories matter. So does the **form** they come in. Sugary drinks are the easiest calories to overdo.

Myth #10: "The sugar industry shaped the science."

VERDICT: TRUE (for a well-documented case in the 1960s)

What the evidence says: In 2016, researchers at UC San Francisco published internal sugar industry documents in *JAMA Internal Medicine* [26]. Here is what they found:

- In 1965, the **Sugar Research Foundation** paid for a research review called "Project 226."
- The review was published in the *New England Journal of Medicine* in 1967.
- It **singled out fat and cholesterol** as the dietary causes of heart disease and **downplayed evidence about sugar**.
- The Sugar Research Foundation set the review's goal, sent articles to include, and saw drafts.
- **The industry's funding and role were not disclosed** [26].

The authors concluded the industry ran a research program in the 1960s and 1970s that "successfully cast doubt about the hazards of sucrose while promoting fat as the dietary culprit" [26].

The nuance: This is a documented case from more than 50 years ago. It does not mean all modern sugar research is fake. Today, journals require authors to disclose funding. Read those disclosures. The authors themselves recommend giving less weight to food industry–funded studies [26].

Evidence strength: Strong. It's based on the industry's own documents.

Bottom line: It happened. It's a good reason to check who paid for a study.

Myth #11: "Artificial sweeteners are a safe, smart swap."

VERDICT: IT'S COMPLICATED

There are really **two** questions here: **Are they safe?** and **Do they help?**

1. Safety: approved, and widely studied.

- The **FDA** says sweeteners it has approved are safe for the general population under approved conditions of use. It calls aspartame "one of the most studied food additives in the human food supply" [29].
- In July 2023, the WHO's cancer agency (**IARC**) placed aspartame in **Group 2B, "possibly carcinogenic to humans,"** based on **limited** evidence [28]. That is the third of four levels. It reflects how strong the evidence is, **not** how risky a normal amount is [28].
- At the same time, a separate WHO/FAO expert committee (**JECFA**) kept the safe daily limit at **40 mg per kg of body weight**. Its example: a 70 kg (154 lb) adult would need to drink **more than 9 to 14 cans** of diet soda a day to go over that limit, assuming no other aspartame [28].
- The **FDA disagreed** with IARC's classification and pointed to "significant shortcomings" in the studies IARC used [29].
- **Important exception:** people with **phenylketonuria (PKU)**, a rare genetic condition, should avoid or limit aspartame [29].

2. Do they help with weight? Mixed evidence.

- In May 2023, the **WHO advised against** using non-sugar sweeteners to control weight or lower disease risk. It found **no long-term benefit** for body fat. This is a **conditional** recommendation, because the studies linking sweeteners to disease may be confounded. It does **not** apply to people who already have diabetes [27].
- But in an 18-month randomized trial in kids, swapping one 8 oz sugary drink a day for a sugar-free one led to **less** weight gain and fat gain [31].

- The new **2025–2030 Dietary Guidelines** say to **limit** low-calorie non-nutritive sweeteners and that no amount is recommended as part of a healthy diet [3].

Evidence strength: Safety reviews are extensive. Long-term benefit for weight is unsettled.

Bottom line: A diet soda is not a health food. In trials, kids given sugar-free drinks gained less weight than kids given sugary ones [31], and adults given diet cola built up less liver fat than adults given regular cola [36]. **Water is better than both.** If you have diabetes, ask your doctor or dietitian what's right for you.

Myth #12: "Juice is a health drink."

VERDICT: MISLEADING

What the evidence says:

- The **WHO** counts the sugar in fruit juice as "free sugar," right alongside table sugar and honey [10].
- The **FDA** counts sugar from **concentrated** fruit juice added to foods as added sugar [1].
- The **American Academy of Pediatrics** says fruit juice "offers no nutritional benefits over whole fruit" for kids [23]. It recommends [23]:
 - **No juice** before 12 months of age
 - Ages 1–3: **4 ounces or less** a day
 - Ages 4–6: **4 to 6 ounces or less** a day
 - Ages 7–18: **8 ounces** a day
- In the large Harvard study, fruit juice was linked to a **higher** risk of type 2 diabetes, while whole fruit was linked to a lower risk [22].
- The CDC lists a "fruit juice drink" at **10 teaspoons** of sugar per 12 ounces [6].

The grain of truth: 100% juice does have vitamins. The 2025–2030 Guidelines allow it in **limited portions or diluted with water** [3]. "Fruit **drinks**" (punches, "-ades," cocktails) are not juice. They are sugary drinks.

Bottom line: Juice is not a health drink. It is a treat in a small glass. **Eat the orange. Drink water.**

Myth Verdict Summary

#	Claim	Verdict
1	Sugar is as addictive as cocaine	Misleading
2	Sugar makes kids hyper	False (for most kids)
3	Sugar directly causes diabetes	It's complicated
4	Honey, agave, and coconut sugar are healthy sugars	False
5	Fruit is just as bad as candy	False
6	HFCS is way worse than table sugar	Misleading
7	Fat-free foods are the healthier choice	Misleading
8	Brown sugar is healthier than white	False
9	A calorie is a calorie, so sugar doesn't matter	Misleading
10	The sugar industry shaped the science	True (1960s, documented)
11	Artificial sweeteners are a safe, smart swap	It's complicated
12	Juice is a health drink	Misleading

Part 6: What the Research Says About Added Sugar and Health

Let's be clear about what we can and can't say. Most of this research is **observational**. It shows that people who eat more added sugar tend to have more of certain health problems. That is a **link**, not proof. Where we have **randomized trials**, we say so.

A big 2023 *BMJ* "umbrella review" pulled together **73 meta-analyses** covering **83 health outcomes**. It found that eating a lot of sugar was generally "more harmful than beneficial," especially for heart and metabolic health. But it rated most of the evidence as **low quality**. The strongest (moderate-quality) findings were for **body weight** (from sugary drinks) and **fat building up in organs like the liver** (from added sugars) [38].

Weight

What we know:

- A 2013 *BMJ* review found that in adults eating freely, **cutting sugar was linked to a small weight drop** (about 0.8 kg, under 2 pounds) and adding sugar to a similar gain. Swapping sugar for the same calories from other carbs made no difference [30]. So the effect runs mostly through calories.
- In kids, the same review found those drinking the most sugary drinks had **55% higher odds** of being overweight a year later than those drinking the least [30].
- In a randomized trial in the Netherlands, 641 kids got either a sugary or a sugar-free 8 oz drink every day for 18 months. Both groups grew, as kids do. The sugary-drink group gained about **1 kg more** [31].
- In a U.S. trial with teens, cutting sugary drinks for a year led to less weight gain at 1 year, but the difference had faded by year 2 [32].

Evidence strength: Moderate to strong, especially for sugary drinks.

Tooth decay

This is one of the clearest links. Bacteria in your mouth feed on sugar and make acid that damages teeth.

- A systematic review done for the WHO found 42 of 50 studies in children and all 5 studies in adults showed at least one link between sugar and cavities [33].
- Cavities were **lower** when free sugars were **under 10%** of calories (moderate-quality evidence) [33] [10].

Evidence strength: Moderate. Consistent across many studies.

Type 2 diabetes

- Sugary drinks are consistently **linked to** higher type 2 diabetes risk: **26% higher** for the heaviest drinkers [20], and **18% higher per daily serving** (13% after accounting for weight) [21].
- In the U.S., one 2015 estimate suggested about **1.8 million** of 20.9 million new type 2 diabetes cases over 10 years could be **attributable** to sugary drinks, under specific assumptions [21].

Evidence strength: Strong for the link with sugary drinks. Observational.

Heart disease

- A study led by CDC researchers, published in *JAMA Internal Medicine* in 2014, followed U.S. adults for a median of **14.6 years** [34].
- People who got **10% to 24.9%** of calories from added sugar had about a **30% higher risk of dying from heart disease** than those under 10%.
- People who got **25% or more** of calories from added sugar had about **2.75 times** the risk [34].
- In the same study, **most U.S. adults (71.4%)** got 10% or more of calories from added sugar (2005–2010 data) [34].

A 2019 review concluded that cohort studies show a strong link between sugary drinks and coronary heart disease, and that short-term trials on risk factors back this up [35]. The AHA's 2016 statement on kids found **strong evidence** linking added sugars with higher heart risk factors in children through extra calories, more body fat, and unhealthy blood fats [9].

Evidence strength: Consistent, but mostly observational. People who eat lots of added sugar may differ in other ways too, even after researchers adjust for it.

Fatty liver

Your liver processes fructose, one half of table sugar and HFCS. Eating too much is linked to fat building up in the liver [37].

- In a 6-month randomized trial, 47 adults with overweight drank **1 liter a day** (about 34 oz) of regular cola, milk, diet cola, or water. Liver fat rose much more in the regular cola group than in the other groups. Total body fat did not differ [36]. This was a small trial with a lot of soda.
- A 2018 review in the *Journal of Hepatology* called sugar a major driver of non-alcoholic fatty liver disease and said early studies suggest cutting sugary drinks may help lower liver fat. It also called for larger trials [37].
- The 2023 umbrella review rated the link between added sugars and fat in organs as **moderate quality** [38].

Evidence strength: Moderate. Trials are small, but they point the same way as the observational data.

The honest summary: No single food decides your health. But across weight, teeth, diabetes, heart, and liver, **too much added sugar, especially in drinks, keeps showing up on the wrong side of the scale.** Cutting back is one of the simplest, lowest-risk changes a family can make.

Part 7: Read a Label in 30 Seconds

You don't need a nutrition degree. You need five steps.

Step	Time	Look at	Ask yourself
1	5 sec	Serving size + servings per container	Will I eat one serving, or the whole thing?
2	5 sec	Includes Added Sugars (grams)	How many grams in what I'll actually eat?
3	5 sec	% Daily Value for Added Sugars	5% or less = low. 20% or more = high [1].
4	5 sec	Do the teaspoon math	Grams $\div$ 4 = teaspoons.
5	10 sec	Ingredient list	Sugar, syrup, "-ose," honey, nectar, or juice concentrate near the top?

Let's practice: the 20-ounce soda

- **Serving size:** 1 bottle (20 fl oz). Servings per container: 1 [11].
- **Added Sugars:** 65 g [11].
- **%DV:** 130%. That is way past "high" [11].
- **Teaspoons:** $65 \div 4 =$ **about 16 teaspoons**.
- **Compared to the 2025–2030 benchmark:** 65 g is **6½ times** the 10 g-per-meal limit [3].

Watch the serving size trap. If a bag says 8 g added sugar per serving and has 3 servings, eating the whole bag is **24 g**. That's almost the AHA's full-day limit for women and kids (25 g) [7][9].

Quick label benchmarks

If you see...	Think...
0 g Added Sugars	Great. (Natural sugar in plain milk or fruit is fine.)
2.5 g or less in a ⅔-cup yogurt	Meets the new dairy snack limit [3].
5 g or less in a ¾-oz whole-grain snack	Meets the new grain snack limit [3].
10 g or less for a whole meal	Meets the 2025–2030 per-meal limit [3].
20% DV or more (10 g+) in one serving	High in added sugar [1].

Part 8: Practical Swaps for Real Families

No crash diets. No banned-food lists. Just trades that add up. Pick one or two to start.

Start with drinks (the #1 source)

Sugary drinks are the leading source of added sugar in the American diet [6]. That makes them the best place to start.

Instead of...	Try...
Regular soda (12 oz: ~10 tsp) [6]	Plain sparkling water with lemon, lime, or cucumber [6]
Fruit juice drink (12 oz: ~10 tsp) [6]	Water with berries, or a splash of 100% juice in sparkling water [6]
Sweet tea (12 oz: ~7 tsp) [6]	Unsweetened iced tea (0 tsp) [6]
Sports drink for everyday sipping (12 oz: ~5 tsp) [6]	Water. Keep a cold jug in the fridge [6].
Energy drink (12 oz: ~9 tsp) [6]	Water, unsweetened tea, or black coffee (for adults) [6]
Flavored milk	Plain milk [6]

Coffee shop tip: The CDC says to skip the flavored syrups and whipped cream [6]. Before you order, look up your usual drink on the chain's nutrition page. Check the **size**. Going from the largest to the smallest size is one of the fastest swaps there is.

Breakfast

Instead of...	Try...
Flavored yogurt	Plain yogurt + your own fresh fruit. Aim for ≤ 2.5 g added sugar per $\frac{2}{3}$ cup [3].
Frosted or sweetened cereal	Check the label. Pick one with low added sugar per serving and measure your bowl.
Pastries or doughnuts	Eggs, whole-grain toast, oatmeal with fruit
Pancakes swimming in syrup	Smaller pour, or top with fruit instead

Snacks and desserts

Instead of...	Try...
Fruit snacks or gummies	Whole fruit: apple slices, grapes, a banana, berries
Candy every day	Candy on special occasions. Smaller portions.
Packaged cookies	Nuts, cheese, popcorn, or a small piece of dark chocolate
"Fat-free" sweet snacks	Read Added Sugars , not the front of the box [24]

A simple 4-week family plan

Go slow. One change a week. Keep what works.

Week	Focus	One simple action
1	Know your number	Read the Added Sugars line on 5 things you eat or drink every week. Write them down.
2	Fix the drinks	Replace one sugary drink a day with water or sparkling water. Fill a reusable bottle [6].
3	Fix breakfast	Swap to plain yogurt + fruit, or a lower-sugar cereal, 3+ days this week.
4	Set the house rules	Stop stocking sugary drinks at home [6]. Save sweets for real occasions.

For parents: Kids copy what you do. The CDC suggests being a role model by choosing water [6]. Don't give toddlers juice in bottles or sippy cups they can sip all day, or at bedtime [23].

Reminder: If you have diabetes, take medication that lowers blood sugar, are pregnant, or have any health condition, talk with your doctor or a registered dietitian before making changes.

Cheat Sheet: Added Sugar at a Glance

THE BASICS

Added sugar	Sugar put in during processing or prep: table sugar, syrups, honey, juice concentrate [1]
Not added sugar	Sugar naturally in whole fruit, vegetables, and plain milk [1]
Label line	"Includes Xg Added Sugars," under Total Sugars [1]
Daily Value	50 g. 5% DV or less = low; 20% DV or more = high [1]
Quick math	Grams ÷ 4 = teaspoons [7][29]

THE LIMITS

Grams / Teaspoons

U.S. Dietary Guidelines 2025–2030	No amount recommended; ≤10 g (≈2½ tsp) per meal [3]
Kids birth–10 (DGA 2025–2030)	Avoid / none recommended [3]
AHA, most women	≤25 g / 6 tsp a day [7]
AHA, most men	≤36 g / 9 tsp a day [7]
AHA, kids 2–18	≤25 g / 6 tsp a day; ≤1 sugary drink (8 oz) a week [9]
WHO	<10% of calories; ideally <5% (≈25 g / 6 tsp) [10]

THE REALITY

Average U.S. adult	17 tsp a day [5]
Average U.S. kid/teen (2–19)	17 tsp a day [5]
#1 source	Sugary drinks: 24% of all added sugar [4]
12 oz regular soda	~10 tsp [6][11]
20 oz Coca-Cola	65 g (≈16 tsp), 130% DV [11]

SPOT THE ALIASES

Words with "sugar" or "syrup"	Cane sugar, brown sugar, HFCS, corn syrup, rice syrup, maple syrup, agave syrup [3]
Words ending in "-ose"	Sucrose, glucose, dextrose, fructose, maltose [3]
"Natural" sweeteners	Honey, molasses, agave nectar, coconut sugar, fruit juice concentrate [3][7]

THE 5 BEST MOVES

1. Drink water	Plain, sparkling, or with fruit slices [6]
2. Read Added Sugars	Check the grams and the serving size [1]
3. Eat whole fruit	Not juice, not fruit snacks [22][23]
4. Go plain	Plain yogurt, plain milk, unsweetened tea. Add your own fruit.
5. Shrink the treat	Smaller sizes, less often [4]

Not medical advice. Talk to your doctor or a registered dietitian, especially if you have diabetes or take glucose-lowering medication.

LoadLine Fitness's **Sugar Smart November** challenge runs **Nov 1–30, 2026** at **noexcuseschallenges.com**.

Read, download, and share this paper free at <https://www.loadlinefitness.com/papers>.

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